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Detecting breast cancer from blood samples

Intermediate Report

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1 Introduction

This work develops a Machine Learning model capable of accurately identifying individuals diagnosed with breast cancer based on a blood sample dataset.
The dataset used in this study consists of data from 116 patients, including 64 diagnosed with breast cancer and 52 belonging to a healthy control group. The nine analyzed variables comprise parameters such as age, body mass index (BMI), glucose and insulin concentrations, the HOMA index, the levels of leptin, adiponectin, resistin, and MCP-1.

–¿ adicionar descrição do problema e aumentar introdução

2 Methods and Methodologies

2.1 Data Partitioning

The data were initially divided into three subsets: a training set (70%), a test set (15%), and a validation set (15%). However, the test and validation sets were later merged, because during this first stage of the project, no hyperparameter tuning was performed at this stage, invalidating the need of a separate test and validation set. The classifiers were trained using the training set, while the test set is reserved for evaluating the classifiers' performance and generalization ability. This was done using a stratified split, ensuring that the proportion of classes (*Healthy* and *Cancer*) was maintained in both subsets.
After the partitioning, the data was normalized according to $\frac{x-\mu}{\sigma}$, so that each feature had a mean of 0 and a standard deviation of 1, ensuring comparability across variables with different value scales.

2.2 Feature Selection and Reduction

Before applying the classifiers, it is necessary to evaluate the discriminative power of the original variables to identify those most relevant for distinguishing between the classes and reducing redundancies. This step aims to later improve the performance of the classifiers and to reduce the dimensionality of the features.

2.2.1 Kruskal-Wallis Test

The Kruskal-Wallis test is a non-parametric statistical test used to compare two or more independent samples. In this context, it was applied to assess

the statistical relevance of each variable in distinguishing between the *Healthy* and *Cancer* classes. The null hypothesis (H_0) assumes that the distributions of samples from different classes come from the same population, i.e., that the variable has no discriminative power. When the p-value is below 0.05, H_0 is rejected, concluding that the samples come from distinct populations and that the feature has discriminative ability.
The function `stats.kruskal` from the `scipy` library was used to compute the p-value for each feature. The features were ranked according to their score values, allowing the identification of those with the greatest discriminative power between healthy and cancer samples.

2.2.2 ROC Curve and AUC

The ROC (Receiver Operating Characteristic) method is particularly suitable for binary classification problems. The ROC curve represents the true positive rate (TPR) as a function of the false positive rate (FPR) for different decision thresholds. The area under the curve (AUC) provides a quantitative measure of the performance of the classifier.
An AUC value of 1 indicates perfect separation between classes, while a value of 0.5 indicates performance equivalent to random guessing. When the value is below 0.5 we simply switch the variables so that the inverse is the one that counts. Therefore the ROC/AUC analysis allows identifying variables with the highest discriminative capacity.

In this analysis, each clinical feature was individually evaluated by computing its ROC curve against the binary class labels (Healthy and Cancer). The `roc_curve` and `auc` functions from the `scikit-learn` library were used to calculate the False Positive Rate (FPR), True Positive Rate (TPR), and the corresponding AUC for each variable.

The results were plotted using the `Plotly` library, generating individual ROC curves for all features in a grid layout, with the AUC value annotated in each subplot. Finally, the features were ranked according to their AUC values, allowing the identification of those with the greatest discriminative power between healthy and cancer samples.

2.2.3 Correlation Matrix

The correlation matrix is used to assess linear relationships between variables. High correlations indicate redundancy, meaning that two features vary similarly. Analyzing this matrix helps select subsets of less correlated variables, reducing redundancy and improving classifier efficiency.

The correlation matrix of the nine features was computed to assess the degree of linear dependence between variables. This analysis was implemented using the `numpy.corrcoef` function and visualized with `Plotly Express` as a heatmap.

2.2.4 Principal Component Analysis (PCA)

PCA is a dimensionality reduction technique that projects data onto a new set of orthogonal axes, called principal components (PCs), which maximize data variance. The goal is to find a reduced set of variables capturing most of the information contained in the original data with minimal variance loss. Eigenvectors define the direction (axis orientation), while eigenvalues represent the variance explained along those directions. Three criteria were used to determine the optimal number of components to retain:

- **95% Threshold:** select components that collectively explain at least 95% of the total variance;
- **Scree Criterion:** identify the point where the eigenvalue plot (scree plot) stabilizes, indicating components with marginal contribution;
- **Kaiser Criterion:** retain only components with eigenvalues greater than 1 (applicable to normalized data).

The `PCA` class from the `scikit-learn` library was then applied to compute the principal components, eigenvalues, and explained variance ratios.

For the 95% variance criterion, the cumulative explained variance was calculated, and the minimum number of components required to reach 95% of the total variance was identified. Using `Plotly`, a bar chart was created to display the variance explained by each principal component. Additionally a graph was created that shows all the samples projected onto the fist principal component (PC1) The principal components were plotted against their explained variance, along with a threshold line at eigenvalue =1 to apply the Kaiser criterion. Components with eigenvalues greater than one were retained, and the corresponding explained variance was computed. The Scree plot, generated using `Plotly`, was analyzed subjectively to determine the point at which the curve flattens out, indicating where additional principal components contribute little to the total variance.

2.2.5 Linear Discriminant Analysis (LDA)

LDA aims to project data into a lower-dimensional subspace that maximizes class separability while minimizing within-class variability. The method finds a direction (or several) that maximizes the ratio between between-class variance and within-class variance, according to Fisher's criterion. Since the number of discriminant directions that can be found is limited by the number of classes, LDA reduces the data dimensionality to at most *(number of classes − 1)*. The result is a linear combination of the original variables, usable for both visualization and classification. The `LinearDiscriminantAnalysis` class from `scikit-learn` was then fitted to the dataset to compute the optimal discriminant direction that maximizes class separation. The data were projected onto the first linear discriminant (LDA1), and the resulting one-dimensional representation was visualized using a scatter plot, allowing a qualitative evaluation of the separation between healthy and cancer samples.

2.3 Classifiers

After feature selection and reduction, three distance-based classifiers were applied to the data:

2.3.1 Euclidean Distance Classifier

In this classifier, each unknown sample is assigned to the class whose mean is closest in terms of Euclidean distance, defined as:

$$||x - m||_e = \sqrt{\sum_{i=1}^{d}(x_i - m_i)^2}$$

where x represents the sample and m the class mean. This method assumes that all variables have equal weight and similar variance. In the implementation, the mean vector of each class was computed using the selected features, using `np.mean`. Each test sample were then classified by computing its Euclidean distance to both class means and assigning it to the closest class with `np.linalg.norm`. The classifier's performance was evaluated by calculating standard metrics, including sensitivity, specificity, precision, F1-score and accuracy.

2.3.2 Mahalanobis Distance Classifier

The Mahalanobis distance is a more sophisticated metric that accounts for correlations among variables and differences in scale. It is defined as:

$$||x - y||_m = \sqrt{(\mathbf{x} - \mathbf{y})^T \mathbf{C}^{-1} (\mathbf{x} - \mathbf{y})}$$

where $\mathbf{C}$ is the covariance matrix of the variables. This classifier adapts to the true geometry of the data, being more robust when variables are correlated. In the implementation, the mean vector of each class was first computed using the selected features using texttt\np.mean. The pooled covariance matrix of the features was then calculated with texttt\np.var and inverted to account for correlations and differences in scale. Each test sample was classified by computing its Mahalanobis distance to the class means, taking into account the covariance structure, and assigning it to the closest class. Standard performance metrics, including sensitivity, specificity, precision, F1-score and accuracy, were subsequently calculated to evaluate the classifier.

2.3.3 Fisher LDA Classifier

The Fisher LDA classifier combines linear discriminant analysis with the Mahalanobis metric. It determines a weight vector $\mathbf{w}$ that maximizes class separation and defines a decision threshold. Decisions are based on the sign of the linear discriminant:

$$y = \mathbf{w}^T \mathbf{x} + b$$

where b is the bias term. This classifier is particularly effective for two-class problems and when distributions are approximately Gaussian. In the implementation, the mean vectors of each class were first computed for the selected features with texttt\np.mean. The within-class scatter matrices were calculated by summing the outer products of deviations from the class means, and the total within-class scatter matrix was inverted. The Fisher linear discriminant vector $\mathbf{w}$ was then obtained by multiplying the inverse of the within-class scatter with the difference of class means, and the bias b was computed as the negative half of the projection of the mean sum onto $\mathbf{w}$. Each test sample was classified based on the sign of $\mathbf{w}^T\mathbf{x} + b$, and standard metrics, as sensitivity, specificity, precision, F1-score, and accuracy, were calculated to evaluate performance.

2.3.4 ROC-AUC

Additionally to the sensitivity, specificity, precision, F1-score, and accuracy metrics used to evaluate each classifier's performance on the test samples, a ROC-AUC score was calculated using the same methods employed in the feature extraction tests. This metric provides an aggregate measure of the classifier's discriminative ability across all decision thresholds, offering a more comprehensive evaluation of its overall performance.

–¿ estamos a usar o mesmo método usado para as features; estamos usar TPR e FPR pusando roc.curve do scipy para depois calcular AUC

2.4 Experimental Setup

3 First Results

To compare the performance of the classifiers, the metrics were calculated using all features without any preprocessing. The results are presented in the table below.

Table 1: Classifier results using all the features.

Metric	**Euclidean**	**Mahalanobis**	**Fisher LDA**	**ROC-AUC**
Sensitivity (%)	47.37	52.63	52.63	52.63
Specificity (%)	75.00	68.75	68.75	52.63
Precision (%)	69.23	66.67	66.67	52.63
F1-score (%)	56.25	58.82	58.82	52.63
Accuracy (%)	60.00	60.00	60.00	52.63

The ROC curves were plotted, and the AUC values were calculated for each feature, as shown in the figure below.

`ROC_AUC.png`

Figure 1: ROC curves for different feature sets.

According to Figure **??**, the AUC values of all features were ranked in order to identify the top five.

Table 2: Ranking AUC values.

Feature	**p-value**
Feature1	47.37
Feature2	75.00
Feature3	69.23
Feature4	56.25
Feature5	60.00
Feature6	60.00
Feature7	60.00
Feature8	60.00
Feature9	60.00

According to Table **??**, we can conclude that the top five features are:..................... Accordingly, the different classifiers were applied using this method, and the resulting performance metrics for the test data are presented in the table below.

Table 3: Classifier results using the top 5 features (ROC selection).

Metric	**Euclidean**	**Mahalanobis**	**Fisher LDA**	**ROC-AUC**
Sensitivity (%)	47.37	52.63	52.63	52.63
Specificity (%)	75.00	68.75	68.75	52.63
Precision (%)	69.23	66.67	66.67	52.63
F1-score (%)	56.25	58.82	58.82	52.63
Accuracy (%)	60.00	60.00	60.00	52.63

CONCLUSÕES TABELA De acordo com a Tabela **??**, verifica-se que os três classificadores dão o mesmo valor de 60% para a accuracy e que os restantes valores não se distanciam muito, como seria de esperar, uma vez que é tudo projetado a uma dimensão. Verifica-se ainda que o classificador Mahalanobis e Fisher LDA apresentam valores iguais para todas as métricas.

We generated the correlation matrix to analyze how the features vary with respect to each other — that is, to determine whether they are highly or weakly correlated and to assess potential redundancy among them (i.e., whether they convey the same information).

CORRELATION_MATRIX.png

Figure 2: Correlation matrix between features.

From Figure **??**, we can observe that Adiponectin is weakly correlated with the other features, as are Age, MCP-1, BMI, and Resistin. (CONFIRMAR)

Next, we applied the Kruskal–Wallis test to reduce dimensionality and identify the top five features whose p-values are greater than 0.05, indicating that the samples belong to the same population.

Table 4: Ranking Kruskal-Wallis test p-values.

Feature	**p-value**
Feature1	47.37
Feature2	75.00
Feature3	69.23
Feature4	56.25
Feature5	60.00
Feature6	60.00
Feature7	60.00
Feature8	60.00
Feature9	60.00

From Table **??**, we can identify the top five features as:......

The performance metrics on the test data for the classifiers applied to the feature selection using the Kruskal–Wallis test are presented in the table below.

Table 5: Classifier results using the top 5 features (Kruskal-Wallis selection).

Metric	**Euclidean**	**Mahalanobis**	**Fisher LDA**	**ROC-AUC**
Sensitivity (%)	47.37	52.63	52.63	52.63
Specificity (%)	75.00	68.75	68.75	52.63
Precision (%)	69.23	66.67	66.67	52.63
F1-score (%)	56.25	58.82	58.82	52.63
Accuracy (%)	60.00	60.00	60.00	52.63

CONCLUSÕES TABELA

Como podemos observar através da Tabela **??**, os resultados são iguais aos obtidos para a curva roc pois selecioanram as mesmas features, como seria de esperar (ENCONTRAR MAIS JUSTIFICAÇÃO PARA OS RESULTADOS DAREM EXTAMENTE IGUAIS)

To analyze the principal components, the three methods described in Section **??** were applied.

The variance explained by each principal component is shown in the figure below.

PC_variance.png

Figure 3: Explained variance for each principaql component.

The 95% criterion indicated that the first seven principal components are required to retain most of the information and preserve the highest variance in the data.

To visualize the selection of PCs according to the Kaiser criterion and the Scree test, the corresponding plot is shown below.

PCA_Kaiser_Scree.png

Figure 4: Princiapal components projected in a Kaiser and Scree plot.

By observing Figure **??**, it can be seen that, according to the Kaiser criterion, only the first four principal components are selected. In contrast, the Scree method does not allow for a clear conclusion, as no leveling-off of the curve is observed.

The first principal component, which accounts for the largest variance, is shown in the figure below.

PC1.png

Figure 5: First principal component.

Among the three criteria used for PCA, the 95% variance threshold provided the best results, as it retained 7 components while preserving most of the information. The Scree criterion, on the other hand, suggested retaining only 4 components, which would likely result in a loss of information since the cumulative explained variance did not reach 95%. Finally, the Kaiser criterion was inconclusive, as it did not allow a clear observation of a stabilization point in the eigenvalues.

Using the different classifiers, the performance metrics were applied to this method, as presented in the table below.

Table 6: Classifier results using the top 7 principal components from PCA.

Metric	**Euclidean**	**Mahalanobis**	**Fisher LDA**	**ROC-AUC**
Sensitivity (%)	42.11	57.89	52.63	52.63
Specificity (%)	81.25	62.50	62.50	52.63
Precision (%)	72.73	64.71	62.50	52.63
F1-score (%)	53.33	61.11	57.14	52.63
Accuracy (%)	60.00	60.00	57.14	52.63

CONCLUSÕES DA TABELA

With LDA, the data was reduced to a single dimension, as shown in the figure below.

LDA1.png

Figure 6: Linear Discriminant in onde dimension.

By applying the classifiers, the resulting performance metrics are presented in the table below.

Table 7: Classifier results using the linear projection obtained by LDA (1D).

Metric	**Euclidean**	**Mahalanobis**	**Fisher LDA**	**ROC-AUC**
Sensitivity (%)	84.21	84.21	84.21	84.21
Specificity (%)	81.25	81.25	81.25	84.21
Precision (%)	84.21	84.21	84.21	84.21
F1-score (%)	84.21	84.21	84.21	84.21
Accuracy (%)	82.86	82.86	82.86	84.21

CONCLUSÃO DOS VALORES TABELAS

4 Conclusion